Print DLP — Microsoft Purview

Implementation Solution Guide

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Executive Summary — Print DLP (Microsoft Purview)

Every other data-loss control watches the file — email DLP, external-sharing blocks, encryption. None of them sees a user open a sensitive document, press **Print**, and walk out with paper that has no audit trail, no revocation, and no way back. Printing is the quiet blind spot in most data-protection programs, and it is exactly the path a regulator or a breach post-mortem will ask about.

This solution closes it using **Microsoft Purview Endpoint Data Loss Prevention (Endpoint DLP)**: a policy that watches the **Print activity** on onboarded devices and — when a document's content matches your **sensitive information types** or carries a sensitive **sensitivity label** — can **audit**, **warn with a policy tip and allow-with-justification**, or **block** the print, while telling **sanctioned corporate printers** apart from **network, local, and print-to-PDF/XPS** output. It stands the control up end to end using only the **Purview portal**, **Security & Compliance PowerShell**, and **Microsoft Graph** — no third-party agent — and is written for an administrator with no prior Endpoint DLP experience.

What you get

• A plain-language **Solution Overview** of the print exfiltration gap, how the Print activity and **printer groups **work, the complementary** sensitivity-label PRINT usage right**, and the simulate→enforce discipline.

• A **Requirements Guide** with verify / implement-if-missing for every prerequisite (the category Purview requirements — tenant/compliance licensing, DLP roles, S&C PowerShell + Graph, in-scope data, stakeholders, pilot + change control — are referenced, not repeated).

• An **Implementation Guide** of precise, ordered steps: build the detectors, define printer groups, create the Devices-only Print policy **in simulation**, tune false positives down, enforce to a pilot ring, then expand — each artifact named, with pass/fail gates and rollback.

• Five **PowerShell scripts** for posture discovery, safe configuration (`-WhatIf`, simulation-only), policy-shape validation, activity reporting, and endpoint-readiness checks.

• **Testing** and **Troubleshooting** guides for a confident rollout.

What this delivers

Printing of sensitive content is **controlled by policy** — proven in simulation, tuned to a low false-positive rate, enforced from a ring outward, with corporate printing preserved, restricted prints audited and alerted, and the control **defensible and evidenced**.

The deployment model

Assess → **Design** → **Configure** → **Pilot (simulate)** → **Validate** → **Expand** — build the policy in simulation, prove it against real print activity, enforce on a handful of users, then roll out in waves.

An honest note

Endpoint DLP requires **onboarded devices and the right licensing** (typically **E5 / E5 Compliance** — **verify your licensing/tenant**), and a Devices-location policy is inert until onboarding is complete. **Windows** print restriction is the most complete; **macOS** print support and specific behaviours evolve — where behaviour is OS-/tenant-dependent this package flags **"verify against current Microsoft guidance / your tenant"** rather than asserting it. Which data is in scope and what the enforced action must be are **compliance, privacy, and legal determinations**, deferred to those functions; this package delivers the technical method and the evidence, not the determination.

Outcome

A print-DLP control that stops sensitive documents from walking out on paper — configured, piloted in simulation, validated against pass/fail gates, and operable — so the capability is in place and defensible in Microsoft Purview.

1. Solution Overview — Print DLP (Microsoft Purview)

What this solution is

Most data-loss controls watch the *file* — email DLP, external-sharing blocks, encryption. None of them see a user select **Print**, send a sensitive document to a printer, and walk away with paper that leaves the building unlogged and unrecoverable. **Microsoft Purview Endpoint Data Loss Prevention (Endpoint DLP)** closes that path by watching the *print activity* on an **onboarded device**: when a document whose content matches your **sensitive information types** (or carries a sensitive **sensitivity label**) is sent to a printer, the policy can **audit** the print, **warn with a policy tip and allow-with-justification**, or **block** it — and can distinguish **corporate/allowed printers** from **local, network, or universal (print-to-PDF/XPS)** printing.

This solution is a repeatable, evidence-driven way to configure the **Print** restriction for sensitive content on endpoints using only the **Microsoft Purview portal**, **Security & Compliance PowerShell** (`Connect-IPPSSession`), and **Microsoft Graph** — no third-party agent. Its defining discipline is that **every restriction is built and proven in SIMULATION / audit mode first** (Microsoft's built-in test mode that records what *would* have been restricted against real device activity, with zero user impact), tuned for a low false-positive rate, and only then moved to enforcement.

> ***Scope boundary.*** *This solution delivers the **technical method and evidence** for restricting printing of sensitive content on endpoints. *Which* data is regulated, *which* classifiers or labels legally apply, and *what* the enforced action must be for a given data category are **business, compliance, privacy, and legal determinations** — deferred to those functions. The solution configures and proves the control; it does not make the policy decision.*

Why this matters

- **Printing is the file controls' blind spot.** A document can be encrypted, un-shareable, and blocked from

email, yet a user opens it and prints it — and none of the file-level controls ever see it. Paper has no audit trail, no revocation, and no DLP at the mail gateway. Endpoint DLP watches the *print activity*, not just the file.

• **Endpoint DLP is licensing- and onboarding-dependent.** The Print restriction requires the right plan (typically **Microsoft 365 E5** / the **E5 Compliance** add-on) and devices **onboarded** to Purview (shared with Microsoft Defender for Endpoint onboarding). A Devices-location policy is inert until onboarding is complete — **verify your licensing/tenant**.

• **Print controls are OS- and printer-dependent.** Print restriction is most complete on **onboarded Windows 10/11 **devices;** macOS **print support and specific behaviours differ and evolve — this solution flags "verify your licensing/tenant" rather than asserting availability.** Printer groups** (corporate vs network vs local vs universal/PDF) let you allow sanctioned printing while restricting the rest.

• **Enforced-first is the classic failure.** Blocking printing straight away breaks legitimate work — a clerk printing a routine form, a nurse printing a chart at the ward printer — floods the help desk, and gets the policy disabled. Audit/simulation first, tuned, is the difference between a control that survives and one that is turned off.

How it works

1. **Sensitive information types (SITs), classifiers, and sensitivity labels** define *what* content is sensitive. Endpoint DLP matches the content of the document being printed against built-in SITs (credit card, national ID, health identifiers, etc.), **custom SITs**, **Exact Data Match (EDM)** SITs, **trainable classifiers**, or the presence of a **sensitivity label** on the item.

1. **A DLP policy scoped to the Devices (Endpoint) location** binds those detectors to the **Print** endpoint activity under *"Audit or restrict activities on devices."* The action per activity is **Audit**, **Block with override**, or **Block**, and can differentiate **printer types** using **printer groups**.

1. **Printer groups** (defined once in **Endpoint DLP settings**) let you treat classes of printers differently — e.g. allow printing sensitive content to a locked-down **corporate** printer group while auditing or blocking **network**, **local (USB/directly attached)**, and **universal/print-to-PDF/XPS** printing.

1. **Simulation / audit mode** runs the policy against live device activity and records what *would* have been restricted — with or without policy tips shown — so you see the true volume and false positives **before** any enforcement. Every restriction in this solution starts here.

1. **Tuning** reduces false positives using confidence levels, instance thresholds, supporting keywords, EDM, label-based conditions, and **printer-group / app exceptions**, measured against the simulation results.

1. **Enforcement** turns the tuned restriction on (tip / allow-with-justification / block) in a limited ring, then expands.

1. **Incident/alert handling** — restricted print activities raise **alerts** in the Purview portal and flow to **Activity Explorer**; this solution wires the triage, severity, and response workflow so matches are actually acted on.

The activity this solution controls (and what it stops)

• **Print** → sending a document that contains sensitive content to any printer. Endpoint DLP can **audit**

the print, **warn-and-justify**, or **block** it, and can apply **different actions per printer group** — the core lever for allowing sanctioned corporate printing while restricting network, local, and print-to-PDF/XPS output. **Availability of specific print behaviours and macOS support varies — verify your licensing/tenant.**

A complementary lever: sensitivity-label protection (RMS usage rights)

Where content carries a **sensitivity label with encryption**, the label's **usage rights** independently govern printing: removing the **PRINT** usage right from a permission set prevents printing protected Office files regardless of endpoint. This solution's primary path is **Endpoint DLP** (which covers any content, labelled or not, and adds printer-group differentiation), but the label-usage-right lever is documented here as a complementary control where labels are already in use — verify your labeling design with the information-protection owner.

The simulate→enforce journey (the core discipline)

> *Confirm SITs/labels → define **printer groups** (corporate/allowed vs network/local/universal) → build the Endpoint DLP policy with the **Print** activity in **SIMULATION / audit** (no user impact) → measure volume and false positives in **Activity Explorer / DLP alerts** → **tune** (confidence, instance count, keywords, EDM, label condition, printer-group exceptions) → re-simulate until false positives are acceptable → enable with **policy tips + allow-override/justification** in a **pilot ring** → validate → expand to **enforce (block)** across the intended scope → operate (triage alerts, review overrides, re-tune).*

What you can control

- Which **SITs/classifiers/labels** trigger the restriction, and at what **confidence** and **instance count**.
- The **action** on the Print activity (audit → tip → justify → block), and **different actions per printer group** — allow corporate, restrict network/local/universal.
- Which **users / devices / groups** are **included or excluded**, and the **pilot ring** enforced first.
- The **printer groups** and any **app exceptions** so sanctioned printing isn't caught.
- Whether users get **policy tips**, can **override** with justification, and what the override says.

Limitations (be honest with buyers)

- **Endpoint DLP requires onboarded devices and the appropriate licensing** (verify your plan — endpoint DLP is gated behind E5 / the E5 Compliance add-on). A Devices-location policy is inert until onboarding is complete.
- **Print behaviours are OS- and printer-dependent and evolve.** Restriction is most complete on onboarded **Windows 10/11**; **macOS** print support and specific behaviours differ — this solution flags "verify your licensing/tenant" rather than asserting availability.
- Endpoint DLP restricts content it can **classify** (SIT / label match). Content below the

confidence/instance threshold, or that the device agent cannot classify at the moment of the print, will not match. DLP is one control among several, not a guarantee — a determined user can still photograph a screen or retype content.

- The **enforced action for a data category is a compliance/legal decision**, not a technical default — this

solution configures and proves the control; the determination is deferred.

Naming convention (used in every doc and script)

Every artifact this solution creates uses `<Purpose> – <Scope> – <Stage>`, reused verbatim across the docs, scripts, and the Purview portal. The standard set:

Artifact	Name
Endpoint DLP policy (simulation)	`DLP – Endpoint Print – Simulation`
Content rule on that policy	`DLP – Endpoint Print – Rule`
Enforced pilot policy	`DLP – Endpoint Print – Enforce Pilot`
Pilot user/device ring	`DLP Pilot Ring – Stage1`
Corporate/allowed printer group	`Printers – Corporate – Allowed`
Policy-tip / override text	`DLP Notice – Print – Standard`
Incident/alert workflow	`DLP Alerts – Print – Operations`
(Optional) label restricting print	`Confidential – No-Print – Endpoint`

The standard deployment model

> ***Assess*** *current endpoint exposure and classification intent →* ***Design*** *the SIT/label detectors, the printer groups, and the print restriction →* ***Configure*** *the Endpoint DLP policy* ***in simulation/audit*** *→* ***Pilot*** *(simulate → tune → enforce to a ring) →* ***Validate*** *against pass/fail gates (printing of sensitive content restricted, corporate printing allowed, false-positive rate low, incidents handled) →* ***Expand*** *to the full scope and ongoing operations. Every high-impact change is piloted in simulation first, with a documented rollback.*

2. Requirements — Print DLP (Microsoft Purview)

These are the **solution-specific** prerequisites for configuring the **Print** restriction for sensitive content on endpoints. They build on — and do not repeat — the six shared prerequisites in `global-requirements-purview.md` (tenant + compliance licensing, least-privilege Purview roles, S&C PowerShell + Graph toolchain, in-scope data sources, compliance/records/privacy/legal stakeholders, and a pilot scope + change control). Verify each item below before starting the Implementation Guide.

For every requirement: **Description** (what it is), **Why required**, **How to verify**, **How to implement if missing**.

2.1 Licensing sufficient for Endpoint DLP print restrictions

• **Description:** Endpoint DLP activity restrictions — including the **Print** activity — are **plan-gated**,

typically **Microsoft 365 E5** or the **E5 Compliance** add-on. Specific print behaviours (printer-group differentiation, macOS print) may be **newer or vary by OS/tenant** and are gated on minimum OS builds. This extends GR-#1 with the capability-specific plan detail.

• **Why required:** A DLP policy scoped to the Devices location and using the Print activity will not

function without the required plan; the activity may not even be selectable.

• **How to verify:** In the **Microsoft 365 admin center** confirm the subscription/add-ons; in the

Purview portal → **Data Loss Prevention** confirm the **Devices** location and, when authoring a rule, that **Print** appears under *"Audit or restrict activities on devices."* **Flag "verify your licensing/tenant"** — availability and macOS support evolve.

• **How to implement if missing:** Acquire/assign the required licensing (E5 / E5 Compliance) before building

the restriction; if a specific print behaviour is unavailable in your tenant/OS, deliver Windows print restriction first and revisit the rest when the capability is enabled.

2.2 DLP-specific least-privilege roles

• **Description:** The **DLP role groups** in the Purview portal — **Compliance Administrator** or

Compliance Data Administrator to create/manage DLP policies, **DLP Compliance Management** for policy management, **View-Only DLP Compliance Management** for read-only review, and the **Data Loss Prevention** role for alert/incident handling. Global Administrator only where genuinely required. Extends GR-#2.

• **Why required:** Purview enforces fine-grained role groups; policy authoring and alert triage require the

matching role, and least privilege keeps the blast radius small.

• **How to verify: Purview portal** → **Settings** → **Roles & scopes** → **Permissions** — confirm your account's

role-group membership includes the DLP roles you need.

• **How to implement if missing:** Assign the least-privilege DLP role group under change control; separate

policy authors from alert triagers where possible.

2.3 Devices onboarded to Purview (the hard gate for any endpoint restriction)

• **Description:** Endpoint DLP monitors activities on **onboarded Windows 10/11 and macOS devices**.

Onboarding is done in **Purview portal** → **Settings** → **Device onboarding** (or via Intune / a script / Group Policy) and is **shared with Microsoft Defender for Endpoint onboarding** — devices already onboarded to Defender for Endpoint are onboarded for Endpoint DLP. Print restriction additionally depends on

minimum OS builds and OS-specific support (Windows print is the most complete; **macOS print support varies — verify your tenant**).

- **Why required:** A policy scoped to the **Devices** location does nothing on a device that is not onboarded

and reporting. Onboarding is the non-negotiable prerequisite for print control.

- **How to verify: Purview portal** → **Settings** → **Device onboarding** → **Devices** shows onboarded devices;

confirm target devices appear and are reporting. Confirm each device meets the minimum OS build documented by Microsoft for the Print activity. Script #5 (`DLPPRINT-EndpointReadiness.ps1`) surfaces the readiness signals.

- **How to implement if missing:** Onboard devices via the Purview device-onboarding package, Intune, or

Group Policy; confirm the minimum OS build and any required AV/agent state per Microsoft docs before scoping the restriction.

2.4 A classification intent — the SITs, classifiers, or labels that mark "sensitive"

- **Description:** A documented view of **which content must not be printed** (e.g. payment card data,

national IDs, health identifiers, source code, or any item carrying a specific **sensitivity label**) mapped to the **detector** that identifies it — built-in SITs where they fit, **custom SITs / EDM / trainable classifiers** for gaps, and/or a **sensitivity-label condition**. Extends GR-#4.

- **Why required:** The restriction keys off *what* is sensitive; a print rule with no precise detector

either catches nothing or catches everything (false-positive flood).

- **How to verify: Purview portal** → **Data classification** → **Classifiers** → **Sensitive info types** lists

built-in and custom SITs; **Information protection** → **Labels** lists sensitivity labels. Confirm the ones you need exist (or are planned) with the data owners.

- **How to implement if missing:** Build the SIT/label inventory (this solution's Discovery script assesses

built-in SIT coverage and any existing endpoint restrictions); plan custom SITs / EDM / a label condition for categories not covered.

2.5 Printer-group and app-exception intent

- **Description:** The definition of **which printers are sanctioned** for sensitive output (e.g. a

locked-down corporate/managed print queue) versus **network**, **local (USB/directly attached)**, and **universal / print-to-PDF/XPS** printing that should be audited or blocked — expressed as **printer groups** — plus any **applications** whose printing is legitimately exempt. Endpoint DLP evaluates the Print activity against **printer groups** and the app lists in **Endpoint DLP settings**.

- **Why required:** Without printer-group differentiation, the restriction catches all legitimate printing to

the sanctioned corporate queue alongside the risky paths, driving false positives and killing adoption.

- **How to verify: Purview portal → Data loss prevention → Endpoint DLP settings → Printer groups**

shows the defined groups; confirm the sanctioned corporate printers are represented before authoring the rule.

- **How to implement if missing:** Capture the sanctioned printer list (name/IP/USB identifiers) from IT,

define the `Printers – Corporate – Allowed` group and any others, and configure the Endpoint DLP settings lists before enforcing.

2.6 A pilot ring, simulation dwell time, and change control (endpoint-specific)

- **Description:** A named **pilot user/device ring**, an agreed **simulation/audit dwell time** (how long

the policy runs in audit before enforcement — long enough to capture representative print activity, e.g. through a monthly close or billing cycle where printing spikes), and the **change-control window** for moving audit → enforce. Extends GR-#6.

- **Why required:** Print blocking is high-impact and highly visible to users; enforcing without an audit

baseline and a limited ring is how the control breaks day-to-day work. The house model mandates **simulation/audit first**, then a ring, then expand.

- **How to verify:** Confirm the pilot ring membership, that the policy is created in **simulation/audit

mode**, and the approval path to enforce.

- **How to implement if missing:** Define the ring and dwell time and agree change control before enabling

any block action.

2.7 An incident/alert triage owner and destination

- **Description:** A named owner (or queue) for **DLP alerts** raised in **Purview portal → Data loss

prevention → Alerts**, plus where restricted-activity events are reviewed (**Activity Explorer**) and, optionally, where they are forwarded (SIEM / Microsoft Sentinel via the Office 365 Management Activity API / Defender connectors). Severity and response actions defined.

- **Why required:** A restricted print event with no one to triage it is not a control. Alerts and Activity

Explorer are where matches surface; someone must own triage, severity, and response.

- **How to verify:** Confirm the triage owner/queue (`DLP Alerts – Print – Operations`), the alert-severity

scheme, and any SIEM forwarding is configured.

- **How to implement if missing:** Assign the owner, define severity/response, and (optionally) wire SIEM

forwarding before enforcement generates real alerts.

Requirements summary

#	Requirement	Gate
GR 1–6	Category prerequisites (`global-requirements-purview.md`)	Tenant/licensing, roles, toolchain, scope, stakeholders, pilot + change control
2.1	Endpoint DLP print licensing (E5 / E5 Compliance; verify OS/tenant support)	Print activity available/selectable
2.2	DLP least-privilege roles	Author + triage roles assigned
2.3	Devices onboarded to Purview (+ min OS build)	Endpoint location & Print activity functional
2.4	Classification intent — SITs / classifiers / labels	Restriction keys off the correct detector
2.5	Printer groups + app exceptions	Corporate printing allowed; risky paths restricted
2.6	Pilot ring + simulation dwell + change control	Simulate/audit-first enforced
2.7	Alert/incident triage owner + destination	Restricted prints are acted on

> ***Deferred to compliance/privacy/legal:*** *which data categories are in scope, which classifiers/labels legally apply, the enforced action per category, and any regulatory applicability. This solution provides the technical configuration and evidence, not the determination.*

Global Requirements — Microsoft Purview Solutions

These requirements apply to **every** solution in the **Purview** category (implementing and managing Microsoft Purview, sensitivity & retention labels, DLP, data lifecycle & records, insider risk, communication compliance, audit, data security posture management, and data governance). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Purview solution).*** *The **Microsoft Purview portal** (purview.microsoft.com), the **Microsoft 365 admin center**, **Security & Compliance PowerShell** (`Connect-IPPSSession`), and **Microsoft Graph** are authored at exact, current click-by-click / cmdlet detail from Microsoft's public documentation. Microsoft Purview features are heavily **licensing-/plan-dependent** (many require **Microsoft 365 E5** or the **E5 Compliance** add-on; some are in preview) and the portal is **actively evolving** — capability availability and portal paths are **cited or flagged "verify your licensing/tenant"**, never asserted or invented. **Regulatory, records-retention, and legal determinations** (retention schedules, records file plans, legal holds, insider-risk case actions, regulatory applicability) are **deferred to the organization's compliance, records, privacy, and legal functions** — the solutions provide the technical configuration method and evidence, not the determination. Automation is **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets.*

1. Microsoft 365 tenant with the required Purview/compliance licensing

- **Description:** A Microsoft 365 tenant with the **compliance licensing** the capability requires

(E3 for baseline; **E5 / E5 Compliance** for advanced DLP, auto-labeling, insider risk, communication compliance, Audit Premium, DSPM, etc.).

- **Why required:** Most Purview capabilities are plan-gated; the feature won't be available without the

right plan.

- **How to verify:** In the M365 admin center confirm the subscription/add-ons; check the feature is

visible in the Purview portal.

- **How to implement if missing:** Assign/purchase the required licensing before starting.

2. Purview roles (least-privilege) and portal access

- **Description:** Access with the least-privilege **Purview / compliance role** the solution names

(e.g. Compliance Administrator, Compliance Data Administrator, Information Protection, Records Management, Insider Risk, eDiscovery, Audit roles), with Global Administrator only where required.

- **Why required:** Purview uses fine-grained role groups; configuration requires the matching role.
- **How to verify:** Purview portal → **Roles & scopes / Permissions** — confirm the role membership.
- **How to implement if missing:** Assign the least-privilege role group under change control.

3. Security & Compliance PowerShell + Graph toolchain

- **Description:** A workstation with **Security & Compliance PowerShell** (`Connect-IPPSSession` via the

ExchangeOnlineManagement module) and **Microsoft Graph PowerShell**, modern-auth / MFA-capable admin.

- **Why required:** Configuration, automation, and the provided scripts run through these.
- **How to verify:** `Connect-IPPSSession` and `Connect-MgGraph` succeed with the admin account.
- **How to implement if missing:** Install the modules and confirm modern-auth sign-in before running steps.

4. Data sources & scope in scope

- **Description:** The **workloads and data sources** in scope (Exchange, SharePoint, OneDrive, Teams,

endpoints/devices, and — for data governance — the broader data estate) and a **data-classification / sensitivity** intent to drive labels, DLP, and retention.

- **Why required:** Purview policies key off the in-scope workloads and the classification scheme.
- **How to verify:** Confirm the in-scope workloads and the classification/sensitivity approach with

stakeholders.

- **How to implement if missing:** Define the scope and classification intent before authoring policies.

5. Compliance, records, privacy & legal stakeholders

• **Description:** Engagement with the **compliance, records-management, privacy, security, and legal**

stakeholders whose policies the configuration must satisfy (retention schedules, records file plans, regulatory obligations, insider-risk and communication-compliance policy, legal hold).

• **Why required:** Purview configuration must align to enterprise policy and legal/records requirements,

not technical defaults; several capabilities have HR/legal sensitivity.

• **How to verify:** Confirm the applicable policies, the records schedule, and named approvers.

• **How to implement if missing:** Identify the owners and capture the requirements before configuring.

6. A pilot scope and change control

• **Description:** A **pilot scope** (users/sites/policies in simulation or a limited ring) to validate

on before tenant-wide enforcement, and a **change-control** process — most Purview policies support a **simulation / test / disabled** mode that MUST be used before enforcement.

• **Why required:** Every solution follows an **Assess → Design → Configure → Pilot (simulate) → Validate

→ Expand (enforce)** model; DLP, labels, retention, and insider-risk changes are high-impact.

• **How to verify:** Confirm a named pilot scope, that simulation/test mode is available, and a change

window/approval path.

• **How to implement if missing:** Define the pilot and agree change control before enforcing anything.

*The deployment model across every Purview solution is the same: **Assess** → **Design** → **Configure** → **Pilot (simulate/test)** → **Validate** → **Expand (enforce)**, with explicit pass/fail gates and a documented rollback. Automation is delivered as **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets. Regulatory/records/legal determinations are deferred to the relevant governance functions.*

3. Implementation Guide — Print DLP (Microsoft Purview)

This is the exact, ordered procedure to configure the **Print** restriction for sensitive content on **onboarded endpoints** — built in **SIMULATION / audit** mode first, tuned, then enforced in a ring, then expanded. Portal paths and cmdlets are from Microsoft's public documentation; **licensing-/plan-dependent features and evolving/preview portal paths (and macOS print support) are flagged "verify your licensing/tenant"** and are never asserted for your tenant. Do the phases in order. Each phase names its artifacts (from the Overview naming table), states a **pass/fail gate**, and gives a **rollback**.

> ***Automation.*** *The five delivered PowerShell tools run through **Security & Compliance PowerShell** (`Connect-IPPSSession`) and **Microsoft Graph**. They are **read-only or** `-WhatIf` **by default**; writes require an explicit `-Commit` switch, and the config helper is **simulation-only** — it never leaves a policy in enforce. No credentials are stored in code.*

> ***Determinations deferred.*** **Which* content categories are in scope, *which* classifiers/labels legally apply, and the *enforced action* per category are* ***compliance/privacy/legal*** *decisions. This guide configures and proves the control; capture those decisions before Phase 4.*

Phase 0 — Connect and baseline (Assess)

1. **Connect the toolchain** (GR-#3). On the admin workstation:

```
Install-Module ExchangeOnlineManagement -Scope CurrentUser -Force   # provides Connect-IPPSSession
Connect-IPPSSession -UserPrincipalName admin@contoso.com            # Security & Compliance PowerShell
Connect-MgGraph -Scopes 'Organization.Read.All','InformationProtectionPolicy.Read' -NoWelcome
```

1. **Run the posture assessment** (delivered script #1, read-only):

```
.\DLPPRINT-Discovery.ps1 -CsvPath .\DLPPRINT-Posture.csv
```

It reports existing DLP policies and their mode, which ones scope the **Devices/Endpoint** location, any **Print** restriction already configured on their rules, and built-in vs custom SIT coverage.

1. **Confirm licensing and endpoint readiness** (Req 2.1, 2.3): confirm E5 / E5 Compliance, that target

devices are **onboarded** and meet the **minimum OS build**, and that the **Print** activity is selectable in a rule (script #5 surfaces the readiness signals). If a specific print behaviour (e.g. macOS print) is not available in your tenant, note it and deliver Windows print first.

Pass/fail gate: you have a written baseline (existing policies + modes + any endpoint restrictions), a SIT/label coverage map, and confirmed licensing + onboarding for the Devices location. **Rollback:** none (read-only).

Phase 1 — Design the classification layer (what counts as "sensitive")

Decide which detector marks content that must not be printed. Do **not** invent SITs — use built-in where they fit; build custom/EDM/trainable only for gaps; use a **sensitivity-label condition** where content is already labelled.

1. **Map built-in SITs first.** In **Purview portal → Data classification → Classifiers → Sensitive info

types**, confirm the built-in SITs for your categories (e.g. *Credit Card Number*, *U.S. Social Security Number (SSN)*, national-ID / health-identifier SITs). Note each SIT's** confidence levels **and default** instance count**.

1. **Author custom SITs for gaps.** For a category with no built-in match (e.g. an internal case or record

number), create a custom SIT with a **regex primary pattern**, **supporting keywords**, a **confidence level**, and a **character proximity**. Portal: **Classifiers → Sensitive info types → Create sensitive info type**. Test it against sample text in the wizard's **Test** step.

1. **Use a sensitivity-label condition where content is labelled.** If the organization already applies a

label such as *Confidential*, the rule can trigger on **"content contains → Sensitivity labels"** instead of (or in addition to) SITs — a precise, low-false-positive signal. Confirm the labels exist with the information-protection owner.

1. **(Optional) EDM / trainable classifiers** for categories where pattern SITs over-match or the category is

conceptual. Verify availability against your plan; *which fields are in scope is the data owner's determination.*

Pass/fail gate: every in-scope "do not print" category maps to a named, tested detector (built-in SIT, custom SIT, EDM, trainable classifier, or sensitivity label) at a chosen confidence/instance count. **Rollback:** delete unused custom SITs (they have no effect until referenced by a policy).

Phase 2 — Define printer groups and configure the restriction in SIMULATION (Configure — no user impact)

First define **printer groups** so sanctioned printing can be treated differently, then create **one Endpoint DLP policy** with a rule that covers the **Print** activity, in **simulation / audit** mode. Simulation/audit records what *would* be restricted against real device activity, with **no** enforcement and (optionally) with policy tips shown.

1. **Define printer groups** (Req 2.5). Portal: **Data loss prevention → Endpoint DLP settings → Printer

groups → **create** `Printers – Corporate – Allowed` **and add the sanctioned corporate printers (by friendly name, IP range, USB product/vendor ID, or "print to local", "network printer", "universal print", "print to file" as your tenant exposes them). This group is what lets you** allow **sensitive printing to the managed queue while** auditing/blocking** network, local, and print-to-PDF/XPS output.

1. **Create the policy —** `DLP – Endpoint Print – Simulation`**.**

Portal: **Data loss prevention → Policies → Create policy →** custom (or a category template) **→ Locations: Devices only** (all other locations off for this solution). Confirm the Devices location is selectable; if not, revisit licensing/onboarding (Req 2.1/2.3).

1. **Author the rule —** `DLP – Endpoint Print – Rule`**.**

• **Condition:** *Content contains* → the SITs from Phase 1 at the chosen confidence/instance count,

and/or *Sensitivity labels* from Phase 1.

• **Actions → "Audit or restrict activities on devices":** set the **Print** activity (labelled per your

tenant's current portal) to **Audit** (the simulation-phase action). Where the portal exposes **per-printer-group** actions, set **corporate/allowed** to *Audit* (later *Allow/Audit only*) and **network / local / universal (print to PDF/XPS)** to *Audit* now. Leave the other endpoint activities (USB, clipboard, network share, etc.) untouched — they are out of scope for this solution.

• **User experience:** define `DLP Notice – Print – Standard` policy-tip text and enable **allow override

with justification** where compliance permits — note tips/overrides only *appear* once the policy is enforced; in audit they are captured, not shown, unless "simulation with policy tips" is selected.

1. **Set the policy mode to simulation.** Choose **"Run the policy in simulation mode"** (optionally *with

policy tips*). Finish. In S&C PowerShell this corresponds to `Mode = TestWithNotifications` / `TestWithoutNotifications`.

1. **Verify with the config helper** (delivered script #2, `-WhatIf` by default):

```
.\DLPPRINT-Config.ps1 -PolicyName 'DLP - Endpoint Print - Simulation' `
    -SensitiveTypes 'Credit Card Number' -MinCount 1 -Confidence High -WhatIf
```

The helper previews the `New-DlpCompliancePolicy` / `New-DlpComplianceRule` calls (Devices location, simulation mode, Print restriction). Because the exact **endpoint-restriction setting names and printer-group syntax evolve**, the recommended pattern is: build the rule **once in the portal**, then read its restriction property with script #1 to confirm the exact names before ever committing with the helper. Run with `-Commit` only to apply an intended change; the helper never leaves a policy in enforce mode.

Pass/fail gate: the policy exists, is scoped to **Devices only**, covers the **Print** activity, treats the `Printers – Corporate – Allowed` group distinctly, and is confirmed in **simulation** (script #1/#2/#3 report a `Test*` mode). **Rollback:** delete the simulation policy — simulation has no user impact, so removal is clean.

Phase 3 — Measure and tune (drive false positives down)

Let the policy run in simulation for the agreed **dwell time** (Req 2.6) to capture representative print activity — ideally spanning a period of peak printing (month-end, billing run).

1. **Review what would have been restricted. Purview portal** → **Data loss prevention** → **Alerts**

(simulation alerts) and **Data classification** → **Activity explorer** (filter to endpoint DLP activities — *print*) show every simulated match with the printer, the SIT/label, and the confidence. Export with the reporting script:

```
.\DLPPRINT-Report.ps1 -Days 14 -CsvPath .\DLPPRINT-Activity.csv
```

1. **Classify each match** as true positive or **false positive** and identify the false-positive drivers

(e.g. routine forms printed to the corporate queue, a document that happens to contain a low-confidence pattern).

1. **Tune** to reduce false positives without losing true positives:

• Raise the **confidence level** (e.g. medium → high) and/or the **instance count** so incidental matches

don't fire.

• Prefer a **sensitivity-label condition** over broad SITs where content is labelled — far fewer false

positives.

• Refine **printer groups**: keep sanctioned corporate printing in the `Printers – Corporate – Allowed`

group at *Audit/Allow*, and focus the restriction on **network / local / universal** printing.

• Introduce **EDM** for categories where pattern SITs over-match.

• Add **supporting keywords / proximity** to custom SITs so context is required.

1. **Re-simulate** after each tuning change and re-measure. Repeat until the false-positive rate meets the

validation gate (see Testing).

Pass/fail gate: measured false-positive rate is at/under the agreed threshold on the simulation sample, with true positives (sensitive-content print) still detected and sanctioned corporate printing not caught. **Rollback:** revert the last tuning change (script #2 `-WhatIf` shows the diff before `-Commit`).

Phase 4 — Enforce in a pilot ring (Pilot)

1. **Scope to the pilot ring.** Set `DLP Pilot Ring – Stage1` (users) and the pilot devices as the

included scope on the policy (all other users/devices excluded for now):

```
.\DLPPRINT-EndpointReadiness.ps1 -PolicyName 'DLP - Endpoint Print - Simulation' `
    -IncludeGroup 'DLP Pilot Ring - Stage1' -WhatIf
```

(The scope helper is delivered as part of the config/remediation tool — `DLPPRINT-Config.ps1 -SetScope`. Run `-WhatIf` first; `-Commit` to apply.)

1. **Move the ring to enforcement.** Change **Policy mode** from *simulation* to **Turn it on / Enforce**.

Start the Print activity at **Block with override (justification)** rather than hard block, so the pilot can report false negatives/positives while retaining a business escape hatch. Keep the `Printers – Corporate – Allowed` group at *Allow/Audit*. (Rename to `DLP – Endpoint Print – Enforce Pilot` if your change process tracks the transition.)

1. **Confirm enforcement is live** for the ring only (script #1 reports mode `Enable` and the scoped

identities; script #4 validates the Devices-only, Print-restriction, printer-group configuration). On a ring device, confirm a test print of clearly-fake sensitive content to a **non-corporate** printer now shows the `DLP Notice – Print – Standard` tip and requires justification, while printing to the corporate queue behaves per policy.

1. **Handle incidents.** Wire `DLP Alerts – Print – Operations`: assign the alert owner/queue (Req 2.7),

set severities, and (optionally) forward to SIEM/Sentinel. Triage a real pilot alert end-to-end to prove the workflow.

Pass/fail gate: the pilot ring is enforced (printing of sensitive content to non-corporate printers restricted, tips shown, overrides captured, corporate printing allowed), matches raise alerts, and the triage workflow produces a dispositioned incident. No enforcement outside the ring. **Rollback:** set the policy back to **simulation mode** (single setting) — instantly removes enforcement with no data change.

Phase 5 — Expand and harden (Enforce → Operate)

1. **Expand scope** from the ring to the full intended population once the pilot gate passes, keeping

documented exclusions.

1. **Escalate the action** per compliance's determination — e.g. move Print (for network/local/universal

groups) from *block with override* to **hard block** where required — one printer group at a time, re-validating after each.

1. **Operationalize:** schedule the reporting script for a recurring false-positive/alert review, review

user overrides (are people justifying legitimately or bypassing?), refresh **EDM** data on cadence (if used), and re-tune the **printer groups** as printers change.

1. **Validate end-to-end** with the delivered testing procedure (next section) and record the pass/fail

evidence.

Pass/fail gate: enforcement live across the intended scope with the compliance-mandated action per printer group; sanctioned corporate printing unaffected; false-positive rate steady/low; alerts triaged on cadence; overrides reviewed. **Rollback:** the policy can be returned to **simulation** at any time as an immediate, non-destructive safety switch; the Print action can be relaxed per printer group independently.

Order-of-operations summary

Phase	Do	Artifact(s)	Gate	Rollback
0 Assess	Connect, baseline, confirm licensing/onboarding	Posture CSV	Baseline + readiness confirmed	n/a (read-only)
1 Design	Map/build SITs / label condition	Custom SIT / label condition	Every category → tested detector	Delete unused SIT
2 Configure	Define printer groups; build the Devices policy (Print) **in simulation**	Printer group + …–`Simulation` policy + rule	Simulation, Devices-only, Print activity, corporate group distinct	Delete sim policy
3 Tune	Measure + reduce false positives	Activity CSV	FP rate ≤ threshold; corporate printing not caught	Revert tuning change
4 Pilot	Enforce to ring, wire alerts	Pilot ring, alert triage	Ring enforced, incident dispositioned	Back to simulation
5 Expand	Broaden scope, escalate to block, operate	Enforced policy	Full-scope enforce, low FP, alerts triaged	Simulation / relax printer group

5. Testing Guide — Print DLP (Microsoft Purview)

Run after the Endpoint DLP print policy is built in **simulation** (Implementation Phases 2–3) and again after each enforcement stage (Phases 4–5). Back each test with `DLPPRINT-Discovery.ps1` (posture), `DLPPRINT-Validation.ps1` (policy shape), `DLPPRINT-EndpointReadiness.ps1` (onboarding), and `DLPPRINT-Report.ps1` (activity). Where print behaviour is **OS-/tenant-dependent (e.g. macOS), verify against current Microsoft guidance / your tenant** rather than assuming.

Test 1 — Licensing and endpoint readiness

Do: Run `DLPPRINT-EndpointReadiness.ps1`; check **Purview portal** → **Settings** → **Device onboarding**. **Expected:** Target devices are **onboarded** and reporting, meet the minimum OS build, and the **Print** activity is selectable when authoring a Devices-location rule. Any gap is listed to fix before enforcement.

Test 2 — Policy is Devices-only and in simulation

Do: Run `DLPPRINT-Discovery.ps1` / `DLPPRINT-Validation.ps1`; check the policy in the portal. **Expected:** `DLP – Endpoint Print – Simulation` exists, scoped to **Devices only**, its rule sets the **Print** activity, and its mode is a `Test*` **(simulation)** mode — not enforce.

Test 3 — Printer groups resolve correctly

Do: Run `DLPPRINT-Validation.ps1`; check **Endpoint DLP settings** → **Printer groups**. **Expected:** `Printers – Corporate – Allowed` exists and contains the sanctioned corporate printers; the rule treats corporate printing distinctly from network / local / universal (print-to-PDF/XPS).

Test 4 — Simulation detects a sensitive print (true positive)

Do: On a ring device, print a document containing clearly-**fake** but SIT-matching content (e.g. a test credit-card / SSN pattern) to a **non-corporate** printer. Run `DLPPRINT-Report.ps1 -Days 1`. **Expected:** The print appears in **Activity Explorer** / simulation **Alerts** as a match, with the SIT, confidence, and printer — showing it *would* have been restricted. No user impact (simulation).

Test 5 — Corporate printing is not caught

Do: Print the same test document to a printer in `Printers – Corporate – Allowed`. Re-run the report. **Expected:** Printing to the sanctioned corporate queue is **allowed / audit-only** per design — not flagged as a would-be block. This is the "don't break legitimate printing" gate.

Test 6 — Non-sensitive print does not match (no false positive)

Do: Print a plainly non-sensitive document (no SIT/label match) to any printer. **Expected:** No print DLP match is recorded — the detector is specific enough not to fire on ordinary content.

Test 7 — False-positive rate meets the gate

Do: After the agreed simulation dwell time, run `DLPPRINT-Report.ps1 -Days <dwell>` and classify matches true/false positive. **Expected:** The measured **false-positive rate is at/under the agreed threshold**, with true positives still detected. If not, tune (Implementation Phase 3) and re-measure before enforcing.

Test 8 — Enforcement blocks with override on the pilot ring

Do: After moving `DLP Pilot Ring – Stage1` to enforce (Block with override), print sensitive test content to a **non-corporate** printer on a ring device. **Expected:** The `DLP Notice – Print – Standard` policy tip appears and the print requires **justification to override** (or is blocked outright if hard block is configured). Corporate printing still works.

Test 9 — Enforcement is scoped to the ring only

Do: Run `DLPPRINT-Discovery.ps1` / `DLPPRINT-Validation.ps1`; on a **non-ring** device print the same sensitive test content. **Expected:** The report shows mode `Enable` scoped to the ring identities only;

the non-ring device is **not** restricted. No enforcement leaks outside the pilot.

Test 10 — Alerts raise and triage end-to-end

Do: From the enforced ring, trigger a restricted print; check **Data loss prevention** → **Alerts** and the `DLP Alerts – Print – Operations` queue. **Expected:** A **DLP alert** is generated at the configured severity, is assigned to the owner/queue, and can be **dispositioned** (and forwarded to SIEM/Sentinel if wired). A restricted print with no triage is not a control.

Pass/fail gate

Gate	Requirement
Readiness pass	Tests 1–3 pass (onboarded + licensed; Devices-only simulation policy; printer groups resolve)
Simulation pass	Tests 4–7 pass (sensitive print detected; corporate printing not caught; no FP on ordinary content; FP rate ≤ threshold)
Enforcement pass	Tests 8–10 pass (ring blocks with override; scoped to ring only; alerts raise and triage)

> *Do not expand beyond the pilot until all three gates pass. Re-run Tests 4–10 after each expansion wave and after any printer-group or detector change.*

6. Troubleshooting Guide — Print DLP (Microsoft Purview)

Symptom → likely root cause → fix. Use `DLPPRINT-Discovery.ps1` (posture), `DLPPRINT-EndpointReadiness.ps1` (onboarding/licensing), `DLPPRINT-Validation.ps1` (policy shape), and `DLPPRINT-Report.ps1` (activity) to confirm before and after each fix. Where print behaviour is **OS-/tenant-dependent (e.g. macOS) or evolving, verify against current Microsoft guidance / your tenant** rather than assuming.

A. Availability / setup

The "Print" activity isn't selectable when authoring a rule

- **Licensing insufficient.** Endpoint DLP activity restrictions are plan-gated (typically E5 / E5

Compliance). **Fix:** confirm the subscription in the M365 admin center; run `DLPPRINT-EndpointReadiness.ps1`. Assign the required licensing (Req 2.1).

- **Devices location not enabled / no onboarded devices.** The Print activity only appears with the Devices

location available. **Fix:** confirm **Settings** → **Device onboarding** shows onboarded devices; onboard targets (Req 2.3).

"Devices" location can't be selected on the policy

- **Onboarding incomplete.** A Devices-location policy is inert until devices are onboarded (shared with

Defender for Endpoint onboarding). **Fix:** onboard via the Purview package / Intune / Group Policy; re-check with `DLPPRINT-EndpointReadiness.ps1`.

B. Nothing is detected (false negatives)

A sensitive print isn't flagged even in simulation

- **Device not onboarded or not reporting.** No onboarded, reporting device = no endpoint signal. **Fix:**

`DLPPRINT-EndpointReadiness.ps1`; confirm the device is onboarded and checked in; confirm the minimum OS build for the Print activity.

- **Detector too strict.** Confidence too high or instance count too large, so real content doesn't match.

Fix: lower confidence/instance count, or switch to a **sensitivity-label condition**; re-simulate.

- **Content not classifiable at print time.** The agent couldn't classify the document (e.g. image-only PDF,

unsupported format). **Fix:** verify the format is supported; consider a label condition for such content.

- **Policy scoped away from the user/device.** The user/device is excluded or outside the included group.

Fix: `DLPPRINT-Validation.ps1` shows the scoped identities; correct the include/exclude.

macOS prints aren't restricted like Windows

- **OS-specific support differs.** Windows print restriction is the most complete; **macOS print support and

specific behaviours evolve**.** Fix: verify against current Microsoft guidance / your tenant** for macOS print support; deliver Windows print first and revisit macOS when the capability is confirmed.

C. Too much is detected (false positives)

Ordinary documents are flagged as would-be blocks

- **Detector too broad.** A low-confidence SIT matches incidental patterns. **Fix:** raise confidence /

instance count, add supporting keywords/proximity, or prefer a **label condition**; re-simulate and re-measure with `DLPPRINT-Report.ps1`.

- **Pattern SIT over-matches structured data. Fix:** introduce **Exact Data Match (EDM)** for

customer/account numbers; refresh EDM data on cadence.

Legitimate corporate printing is being caught

- **Corporate printers not in the allowed group.** The `Printers – Corporate – Allowed` group is missing

printers or the rule doesn't treat it distinctly. **Fix: Endpoint DLP settings** → **Printer groups** — add the sanctioned printers (name/IP/USB IDs); keep that group at *allow/audit* while restricting network/local/universal. Re-run `DLPPRINT-Validation.ps1`.

- **Printer moved / changed IP.** A stale printer group no longer matches the moved corporate printer.

Fix: refresh the printer group whenever the print estate changes.

D. Printer groups

Print-to-PDF/XPS (print to file) isn't being restricted

- **Universal / print-to-file class not covered.** Physical-printer rules don't automatically cover

print-to-file. **Fix:** ensure the **universal / print-to-file** class is in scope of the restriction (not just physical printers) — this is a common quiet exfiltration gap.

A printer lands in the wrong group

- **Ambiguous match (name vs IP vs USB ID).** Overlapping group definitions. **Fix:** tighten the group

membership criteria so each printer resolves to exactly one group; validate with `DLPPRINT-Validation.ps1`.

E. Enforcement / user experience

Users see no policy tip or override prompt

- **Still in simulation.** Tips/overrides only appear once the policy is **enforced** (unless "simulation

with policy tips" is selected). **Fix:** confirm mode with `DLPPRINT-Discovery.ps1` (`Test*` = simulation, `Enable` = enforce); move the ring to enforce when ready.

- **Policy-tip / override not configured. Fix:** set `DLP Notice – Print – Standard` text and enable

allow-override-with-justification in the rule's user-experience settings.

Enforcement leaked outside the pilot ring

- **Scope not limited to the ring. Fix:** `DLPPRINT-Validation.ps1` shows the scoped identities; set the

included scope to `DLP Pilot Ring – Stage1` only (config helper `-SetScope`), all others excluded.

A hard block is stopping essential printing

- **Escalated too fast / wrong printer group. Fix:** immediate rollback — set the policy back to

simulation (single setting, no data change), or relax the affected **printer group** to *block with override* / *audit*; re-tune before re-escalating.

F. Alerts / operations

Restricted prints don't raise alerts (or no one sees them)

- **Alert not enabled / no triage owner. Fix:** confirm the rule's **alert** is on and severity set;

assign the `DLP Alerts – Print – Operations` owner/queue (Req 2.7). A restricted print with no triage is not a control.

- **SIEM forwarding not wired. Fix:** confirm the Office 365 Management Activity API / Defender

connector to Sentinel is configured if forwarding is required.

Reports show rising overrides

- **Detector or printer groups mis-calibrated, or users bypassing. Fix:** review override justifications

in `DLPPRINT-Report.ps1`; if legitimate, calibration is fine; if routine bypass, tighten the detector or printer groups, or escalate the action per compliance.

Escalation checklist

1. `DLPPRINT-EndpointReadiness.ps1` — licensed, devices onboarded/reporting, Print activity available?
2. `DLPPRINT-Discovery.ps1` — policy exists, Devices-only, correct mode (Test* vs Enable), Print restriction set?
3. `DLPPRINT-Validation.ps1` — printer groups resolve, corporate group distinct, scope = intended identities?
4. `DLPPRINT-Report.ps1` — matches, false-positive rate, overrides, restricted printers as expected?
5. Still stuck → capture the print symptom and **verify the current Microsoft guidance** for Endpoint DLP

Print behaviour (especially macOS / printer-group syntax) in your tenant before a fleet-wide change.

7. Best Practices — Print DLP (Microsoft Purview)

Field-proven recommendations for a print restriction that survives contact with real users.

Design

- **Simulate before you enforce — always.** Print blocking is high-impact and visible. Build the policy in

simulation/audit, measure real print volume and false positives, tune, and only then enforce. This is the single biggest predictor of whether the control stays on.

- **Key off precise detectors, not broad ones.** A print rule tied to a **sensitivity label** or a

well-tuned **SIT** (right confidence + instance count) fires on the right documents; a broad, low-confidence SIT floods the help desk and gets the policy disabled. Prefer a **label condition** where content is already labelled.

- **Let compliance own the "what" and "how enforced."** Which categories must not be printed, and whether the

action is audit / warn / block, are **compliance/privacy/legal determinations**. You configure and prove the control; you don't decide the policy.

Printer groups

- **Allow the sanctioned path; restrict the rest.** Define `Printers – Corporate – Allowed` for the managed,

locked-down queue and keep it at *allow/audit*, then focus enforcement on **network**, **local (USB/directly attached)**, and **universal / print-to-PDF/XPS** printing. Blocking *all* printing is almost never the requirement and guarantees pushback.

- **Keep printer groups current.** Printers move, get replaced, and change IPs. A stale group either misses a

new risky printer or wrongly catches a moved corporate one. Review printer groups whenever the print estate changes and re-run `DLPPRINT-Validation.ps1`.

- **Watch print-to-PDF/XPS ("universal").** Printing to a file is a quiet exfiltration path that bypasses

physical-printer controls entirely — make sure the universal/print-to-file class is covered by the restriction, not just physical printers.

Detection & tuning

- **Tune against the simulation sample, one change at a time.** Raise confidence/instance count, add

supporting keywords/proximity, or switch to a label condition — then re-simulate and re-measure. Changing several knobs at once hides which one helped.

- **Use EDM where pattern SITs over-match.** For structured data (customer/account numbers), **Exact Data

Match** dramatically cuts false positives versus a regex SIT. Refresh the EDM data on cadence.

- **Set an honest false-positive threshold and hold to it.** Agree the acceptable rate with the business

before enforcing, and don't move to block until the simulation sample meets it.

Rollout

- **Pilot on a ring, then wave.** Prove enforcement on `DLP Pilot Ring – Stage1` with **block-with-override**

first (a business escape hatch that still captures evidence), pass all three testing gates, then expand in waves — validating each. Never flip the whole tenant at once.

- **Escalate to hard block deliberately.** Move from *block with override* to *hard block* per printer group,

one at a time, only where compliance requires it, re-validating after each. Keep documented exclusions.

- **Simulation is your instant rollback.** Setting the policy back to simulation removes enforcement

immediately with no data change — the safety switch for any surprise after go-live.

Operations

- **Triage means someone, not somewhere.** Assign `DLP Alerts – Print – Operations` to a real owner/queue,

set severities, and (optionally) forward to Sentinel. A restricted print with no triage owner is not a control. Triage a real alert during the pilot to prove the workflow.

• **Review overrides, not just blocks.** If users are justifying overrides legitimately, the policy is

calibrated; if they're bypassing routinely, the detector or printer groups need work. Report on overrides as a first-class signal.

• **Report on a schedule.** Run `DLPPRINT-Report.ps1` on a cadence to watch false-positive drift, new

printers, and alert volume — catch drift before it becomes a ticket storm.

Complementary control

• **Use sensitivity-label PRINT usage rights where labels exist.** For encrypted-labelled Office files,

removing the **PRINT** usage right blocks printing regardless of endpoint or OS — a useful complement to Endpoint DLP for the highest-sensitivity content. Confirm the labeling design with the information-protection owner; don't change usage rights unilaterally.

Automation & security

• **Run the scripts read-only first.** `DLPPRINT-Discovery` / `DLPPRINT-Validation` / `DLPPRINT-Report` /

`DLPPRINT-EndpointReadiness` are read-only; the config helper defaults to `-WhatIf` — preview before `-Commit`, and it never leaves a policy in enforce.

• **Least privilege, no secrets.** Connect with the **least S&C PowerShell / Graph scopes** each task needs;

never place credentials or tokens in scripts or source control.

• **Verify OS-dependent behaviour.** Windows print restriction is the most complete; **macOS print support

and specific behaviours evolve — verify against current Microsoft guidance / your tenant** before relying on them fleet-wide.

8. Deliverables Summary — Print DLP (Microsoft Purview)

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	The print exfiltration gap, how Endpoint DLP's Print activity + printer groups close it, and the Assess→Design→Configure→Pilot(simulate)→Validate→Expand model

Document	Purpose
2. Requirements Guide	Every print-DLP prerequisite with Verify / Implement-if-missing (category Purview GRs referenced, not duplicated): licensing, DLP roles, device onboarding, classification intent, printer groups, pilot ring, alert triage
Global Requirements (Purview)	Category-wide Purview prerequisites, published once
3. Implementation Guide	Exact, ordered Purview portal + S&C PowerShell + Graph steps: detectors, printer groups, the Devices-only Print policy **in simulation**, tune, enforce to a ring, expand — each artifact named, with pass/fail gates + rollback
5. Testing Guide	Validation procedures with pass/fail gates (readiness, simulation, enforcement)
6. Troubleshooting Guide	Print-DLP issues by symptom → root cause → fix
7. Best Practices	Field-proven recommendations for a print restriction that survives real users

PowerShell scripts (Security & Compliance PowerShell / Microsoft Graph)

Script	What it does	Safety
`DLPPRINT-Discovery.ps1`	Posture assessment: existing DLP policies + mode, which scope the Devices/Endpoint location, any Print restriction already configured, built-in vs custom SIT coverage	Read-only
`DLPPRINT-Config.ps1`	Configuration/remediation helper: previews/creates the Devices-only Print policy + rule **in simulation mode**, and (with `-SetScope`) sets the pilot-ring include scope; never leaves a policy in enforce	Default `-WhatIf`; `-Commit` to write
`DLPPRINT-Validation.ps1`	Validates the policy shape: Devices-only, Print activity set, printer groups resolve, simulation vs enforce mode, scoped identities — the "is it built right" report	Read-only
`DLPPRINT-Report.ps1`	Exports print DLP activity (matches, overrides, printers, SIT/label, confidence) over N days from the audit log for false-positive/alert review	Read-only

Script	What it does	Safety
`DLPPRINT-EndpointReadiness.ps1`	Scope/health helper: onboarded-device readiness and licensing signals for the Devices location and the Print activity	Read-only

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process: **Assess** → **Design** → **Configure** → **Pilot (simulate)** → **Validate** → **Expand.**

Outcome

Printing of sensitive documents is **controlled by policy** on onboarded endpoints — proven in **simulation** first, tuned to a low false-positive rate, enforced from a pilot ring outward, with **corporate printing still allowed**, restricted prints **audited and alerted**, and the whole control **defensible and evidenced** in Microsoft Purview. Regulatory/records/legal determinations remain with the relevant governance functions; this package delivers the technical method and the proof.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

DLPPRINT-Config.ps1

Purpose: Builds the Endpoint DLP Print restriction this solution relies on: a DLP policy scoped to the Devices location (all other locations off), in SIMULATION mode, with a rule that triggers on the chosen sensitive-information type(s) and sets the Print endpoint restriction to Audit. With -SetScope it sets the pilot-ring include scope on an existing policy. SAFE BY DEFAULT: previews with -WhatIf; only writes with -Commit; and it NEVER leaves a policy in enforce (mode stays a Test* simulation mode). Idempotent by name. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\DLPPRINT-Config.ps1 -PolicyName 'DLP - Endpoint Print - Simulation' -SensitiveTypes 'Credit Card Number' -WhatIf Create in sim: ... -SensitiveTypes 'Credit Card Number' -MinCount 1 -Confidence High -Commit Set pilot scope: ... -SetScope -IncludeGroup 'DLP Pilot Ring - Stage1' -Commit Requires ExchangeOnlineManagement + a Compliance/DLP admin (Connect-IPPSSession). EXPECTED RESULTS With -WhatIf: prints the New-DlpCompliancePolicy / New-DlpComplianceRule (or Set-*) it WOULD run. With -Commit: the simulation policy + Print rule exist (created if missing), or the include scope is set. Printer-group differentiation and exact restriction names evolve — build the rule once in the portal and confirm names with DLPPRINT-Discovery.ps1 before relying on -Commit. The enforced action per data category is a compliance/legal determination, deferred. .NOTES Purview category — Print DLP (Endpoint DLP). Default -WhatIf; simulation-only (never enforce). No secrets. Verify current Endpoint DLP restriction syntax / printer-group config against Microsoft guidance before -Commit. > [CmdletBinding(SupportsShouldProcess=$true)] param([string]$PolicyName = 'DLP - Endpoint Print - Simulation', [string]$RuleName = 'DLP - Endpoint Print - Rule', [string[]]$SensitiveTypes = @('Credit Card Number'), [int]$MinCount = 1, [ValidateSet('Low','Medium','High')] [string]$Confidence = 'High', [switch]$SetScope, [string]$IncludeGroup, [switch]$Commit)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\DLPPRINT-Config.ps1 -PolicyName 'DLP - Endpoint Print - Simulation' -SensitiveTypes 'Credit Card Number' -WhatIf Create in sim: ... -SensitiveTypes 'Credit Card Number' -MinCount 1 -Confidence High -Commit Set pilot scope: ... -SetScope -IncludeGroup 'DLP Pilot Ring - Stage1' -Commit Requires ExchangeOnlineManagement + a Compliance/DLP admin (Connect-IPPSSession). EXPECTED RESULTS With -WhatIf: prints the New-DlpCompliancePolicy / New-DlpComplianceRule (or Set-*) it WOULD run. With -Commit: the simulation policy + Print rule exist (created if missing), or the include scope is set. Printer-group differentiation and exact restriction names evolve — build the rule once in the portal and confirm names with DLPPRINT-Discovery.ps1 before relying on -Commit. The enforced action per data category is a compliance/legal determination, deferred. .NOTES Purview category — Print DLP (Endpoint DLP). Default -WhatIf; simulation-only (never enforce). No secrets. Verify current Endpoint DLP restriction syntax / printer-group config against Microsoft guidance before -Commit. > [CmdletBinding(SupportsShouldProcess=$true)] param([string]$PolicyName = 'DLP - Endpoint Print - Simulation', [string]$RuleName = 'DLP - Endpoint Print - Rule', [string[]]$SensitiveTypes = @('Credit Card Number'), [int]$MinCount = 1, [ValidateSet('Low','Medium','High')] [string]$Confidence = 'High', [switch]$SetScope, [string]$IncludeGroup, [switch]$Commit)

DLPPRINT-Discovery.ps1

Purpose: Assesses the tenant's readiness for an Endpoint DLP Print restriction. Reports existing DLP policies and their mode (simulation vs enforce), which policies scope the Devices/Endpoint location, whether any rule already configures the Print endpoint restriction, and the built-in vs custom sensitive-information-type inventory. Read-only against Security & Compliance PowerShell — makes no changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\DLPPRINT-Discovery.ps1 -CsvPath .\DLPPRINT-Posture.csv Requires the ExchangeOnlineManagement module and a Compliance/DLP admin. Connect first with Connect-IPPSSession (the script connects if no session is present). EXPECTED RESULTS A per-policy list with mode, whether it scopes Devices, and whether its rules set the Print restriction; a count of built-in vs custom SITs; and (optional) a CSV export. Nothing is modified. Regulatory/records/legal determinations are deferred to the relevant governance functions. .NOTES Purview category — Print DLP (Endpoint DLP). Read-only. Endpoint Print behaviour is OS-/tenant-dependent (Windows most complete; macOS varies) — verify against current Microsoft guidance / your tenant. > [CmdletBinding()] param([string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\DLPPRINT-Discovery.ps1 -CsvPath .\DLPPRINT-Posture.csv Requires the ExchangeOnlineManagement module and a Compliance/DLP admin. Connect first with Connect-IPPSSession (the script connects if no session is present). EXPECTED RESULTS A per-policy list with mode, whether it scopes Devices, and whether its rules set the Print restriction; a count of built-in vs custom SITs; and (optional) a CSV export. Nothing is modified. Regulatory/records/legal determinations are deferred to the relevant governance functions. .NOTES Purview category — Print DLP (Endpoint DLP). Read-only. Endpoint Print behaviour is OS-/tenant-dependent (Windows most complete; macOS varies) — verify against current Microsoft guidance / your tenant. > [CmdletBinding()] param([string]$CsvPath)

DLPPRINT-EndpointReadiness.ps1

Purpose: Reports the readiness signals an Endpoint DLP Print restriction depends on: (1) a licensing signal for Endpoint DLP (typically E5 / E5 Compliance) read from the tenant's subscribed SKUs via Microsoft Graph, and (2) whether the Devices/Endpoint DLP location is usable — i.e. whether any DLP policy can scope Devices — read from Security & Compliance PowerShell. Read-only — makes no changes. Onboarded-device count is confirmed in the portal (Settings → Device onboarding); there is no reliable S&C cmdlet for it, so the script flags it for portal verification rather than inventing one. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\DLPPRINT-EndpointReadiness.ps1 Requires ExchangeOnlineManagement (Connect-IPPSSession) and, for the licensing signal, Microsoft Graph PowerShell (Connect-MgGraph -Scopes 'Organization.Read.All'). The script connects if no session is present; skip -CheckLicensing to run S&C-only. EXPECTED RESULTS A licensing signal (E5 / E5 Compliance service plans present or "verify your licensing/tenant"), confirmation that the Devices DLP location is usable, and an explicit reminder to verify onboarded, reporting devices in the portal before expecting the Print activity to fire. Nothing is modified. .NOTES Purview category — Print DLP (Endpoint DLP). Read-only. Endpoint DLP is licensing-gated and the Print activity is OS-/build-dependent (Windows most complete; macOS varies) — verify against current Microsoft guidance / your tenant. Regulatory/records/legal determinations are deferred to the relevant governance functions. > [CmdletBinding()] param([switch]$CheckLicensing = $true)

How to run: powershell -ExecutionPolicy Bypass -File .\DLPPRINT-EndpointReadiness.ps1 Requires ExchangeOnlineManagement (Connect-IPPSSession) and, for the licensing signal, Microsoft Graph PowerShell (Connect-MgGraph -Scopes 'Organization.Read.All'). The script connects if no session is present; skip -CheckLicensing to run S&C-only. EXPECTED RESULTS A licensing signal (E5 / E5 Compliance service plans present or "verify your licensing/tenant"), confirmation that the Devices DLP

location is usable, and an explicit reminder to verify onboarded, reporting devices in the portal before expecting the Print activity to fire. Nothing is modified. .NOTES Purview category — Print DLP (Endpoint DLP). Read-only. Endpoint DLP is licensing-gated and the Print activity is OS-/build-dependent (Windows most complete; macOS varies) — verify against current Microsoft guidance / your tenant. Regulatory/records/legal determinations are deferred to the relevant governance functions. > [CmdletBinding()] param([switch]$CheckLicensing = $true)

DLPPRINT-Report.ps1

Purpose: Pulls endpoint DLP Print activity from the unified audit log over the last N days so you can measure the simulation/enforcement result: how many prints matched, which sensitive-information types/labels fired, which printers were involved, and how many were overridden. Use it to classify true vs false positives during tuning (Implementation Phase 3) and to watch drift in operations (Phase 5). Read-only — makes no changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\DLPPRINT-Report.ps1 -Days 14 -CsvPath .\DLPPRINT-Activity.csv Requires ExchangeOnlineManagement + audit-log read (Connect-IPPSSession / Connect-ExchangeOnline). Unified audit log must be enabled in the tenant. EXPECTED RESULTS A console summary (total print-DLP events, unique users, unique printers, override count) and an optional CSV of the raw events for classification. If nothing returns, confirm the policy has run long enough and the audit log is enabled. No changes are made. .NOTES Purview category — Print DLP (Endpoint DLP). Read-only. Audit schema evolves — verify field names against current Microsoft guidance / your tenant. > [CmdletBinding()] param([int]$Days = 14, [string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\DLPPRINT-Report.ps1 -Days 14 -CsvPath .\DLPPRINT-Activity.csv Requires ExchangeOnlineManagement + audit-log read (Connect-IPPSSession / Connect-ExchangeOnline). Unified audit log must be enabled in the tenant. EXPECTED RESULTS A console summary (total print-DLP events, unique users, unique printers, override count) and an optional CSV of the raw events for classification. If nothing returns, confirm the policy has run long enough and the audit log is enabled. No changes are made. .NOTES Purview category — Print DLP (Endpoint DLP). Read-only. Audit schema evolves — verify field names against current Microsoft guidance / your tenant. > [CmdletBinding()] param([int]$Days = 14, [string]$CsvPath)

DLPPRINT-Validation.ps1

Purpose: Validates that the Print DLP policy is built correctly: scoped to the Devices/Endpoint location only, its rule sets the Print endpoint restriction, the policy is in the expected mode (simulation vs enforce), and the include/exclude scope matches the intended identities (e.g. the pilot ring). Prints PASS/FAIL per check. Read-only — makes no changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\DLPPRINT-Validation.ps1 -PolicyName 'DLP - Endpoint Print - Simulation' Expect simulation: ... -ExpectMode Simulation Expect ring: ... -ExpectMode Enforce -ExpectScope 'DLP Pilot Ring - Stage1' Requires ExchangeOnlineManagement + a Compliance/DLP reviewer role (Connect-IPPSSession). EXPECTED RESULTS A PASS/FAIL line per check (policy exists, Devices-only, Print restriction present, mode as expected, scope as expected) and an overall result. Any FAIL points at the fix in the Implementation / Troubleshooting guides. Nothing is modified. .NOTES Purview category — Print DLP (Endpoint DLP). Read-only. Printer-group syntax / restriction names evolve — verify against current Microsoft guidance / your tenant. > [CmdletBinding()] param([Parameter(Mandatory=$true)][string]$PolicyName, [ValidateSet('Simulation','Enforce','Any')] [string]$ExpectMode = 'Any', [string]$ExpectScope)

How to run: powershell -ExecutionPolicy Bypass -File .\DLPPRINT-Validation.ps1 -PolicyName 'DLP - Endpoint Print - Simulation' Expect simulation: ... -ExpectMode Simulation Expect ring: ... -ExpectMode Enforce -ExpectScope 'DLP Pilot Ring - Stage1' Requires ExchangeOnlineManagement + a Compliance/DLP reviewer role (Connect-IPPSSession). EXPECTED RESULTS A PASS/FAIL line per check (policy exists, Devices-only, Print restriction present, mode as expected, scope as expected) and an overall result. Any FAIL points at the fix in the Implementation / Troubleshooting guides. Nothing is modified. .NOTES Purview category — Print DLP (Endpoint DLP). Read-only. Printer-group syntax / restriction names evolve — verify against current Microsoft guidance / your tenant. > [CmdletBinding()] param([Parameter(Mandatory=$true)][string]$PolicyName, [ValidateSet('Simulation','Enforce','Any')] [string]$ExpectMode = 'Any', [string]$ExpectScope)